PyMongo Library

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PyMongo

- Documentation: https://pymongo.readthedocs.io/en/stable/
- Source code: https://github.com/mongodb/mongo-python-driver
- Python library of useful tools for interacting with MongoDB database
- Installing
  - pip install pymongo

Connecting with MongoDB Atlas

- Create Cluster and setup connection in MongoDB Atlas (add current IP address or 0.0.0.0/0 to allow access from everywhere)

1 Select your driver and version

DRIVER	VERSION
Python	3.6 or later

2 Add your connection string into your application code

☐ Include full driver code example

```
mongodb+srv://bntan:<password>@cluster1.wte67wg.mongodb.net/?
retryWrites=true&w=majority
```

Replace **<password>** with the password for the **bntan** user. Ensure any option params are URL encoded.

Connecting with MongoDB Atlas

- Connect Colab to MongoDB Atlas by PyMongo

```
import pymongo

client = pymongo.MongoClient("mongodb+srv://bntan:buinhattan@cluster1.wte67wg.mongodb.net/?retryWrites=true&w=majority")
client.list_database_names()
```

```
['CS302', 'admin', 'local']
```

- Create, rename and check collection in database

```
client.CS302.create_collection('Lab01')   #create collection in database
client.CS302.Lab01.rename('Lab04') #rename collection
list(client.CS302.list_collections()) #check the collections in database
```

```
[{'name': 'Lab04',
  'type': 'collection',
  'options': {},
  'info': {'readOnly': False,
   'uuid': Binary(b'a/,\xd0\xa4\xe7E+\x8f\xdas\xdb\xf7v\xa5\x19', 4)},
  'idIndex': {'v': 2, 'key': {'_id': 1}, 'name': '_id_'}},
```

Simple operation

- Insert data item to database collection

```
collection = client.CS302.Lab04

collection.insert_one({'name' : 'nhat-tan bui'})              # Insert item to collection

student_info = {  'name' : "Bui Nhat Tan" ,                          # String
                  'age' : 22,                                        # Integer
                  'gender' : "male",                                 # String
                  'likes_python' : True,                             # Boolean
                  'student_id': {
                      'id' :  "1859043",                             # String
                      'year' : "2018-2022",                          # String
                }}

collection.insert_one(student_info)     # Insert item to collection

list(collection.find())    # Show the item in collection
```

```
[{'_id': ObjectId('639afeca4a8e376909e591f7'), 'name': 'nhat-tan bui'},
 {'_id': ObjectId('639b06264a8e376909e591f9'),
  'name': 'Bui Nhat Tan',
  'age': 22,
  'gender': 'male',
  'likes_python': True,
  'student_id': {'id': '1859043', 'year': '2018-2022'}}]
```

Simple operation

- Find data item in database collection

```
list(collection.find({'name' : "Bui Nhat Tan" }))
```

```
[{'_id': ObjectId('639b06264a8e376909e591f9'),
  'name': 'Bui Nhat Tan',
  'age': 22,
  'gender': 'male',
  'likes_python': True,
  'student_id': {'id': '1859043', 'year': '2018-2022'}}]
```

- Update data item in database collection

```
query = {'name' : "Bui Nhat Tan"}
newvalues = {'$set' : { 'age' : 30 }}

collection.update_one(query, newvalues)
list(collection.find({'name' : "Bui Nhat Tan" }))
```

```
[{'_id': ObjectId('639b06264a8e376909e591f9'),
  'name': 'Bui Nhat Tan',
  'age': 30,
  'gender': 'male',
  'likes_python': True,
  'student_id': {'id': '1859043', 'year': '2018-2022'}}]
```

Simple operation

- Delete data item in database collection

```
collection.delete_one({'name': "nhat-tan bui"})
list(collection.find())
```

```
[{'_id': ObjectId('639b06264a8e376909e591f9'),
  'name': 'Bui Nhat Tan',
  'age': 30,
  'gender': 'male',
  'likes_python': True,
  'student_id': {'id': '1859043', 'year': '2018-2022'}}]
```